

The Operator Carries the Subject: Subject-Aware EEG Denoising by Calibrated Artifact Operators over a Population Diffusion Prior

ANONYMOUS AUTHOR(S)

Ocular artifacts contaminate electroencephalography (EEG) differently in every person, because the propagation of eye potentials to the scalp depends on anatomy and electrode placement. Denoising networks trained on a population absorb this variation into shared parameters and cannot state what should be removed for a new user. We separate the two problems. A conditional diffusion model, trained once on a population, represents the structure of clean EEG. A two-minute calibration segment recorded from each user provides a closed-form estimate of that user's electrooculography (EOG) to EEG propagation matrix, stabilized by empirical Bayes shrinkage toward the population estimate. The propagated EOG then guides the diffusion model as an explicit estimate of the ocular contribution. When a calibration is unreliable the method withholds the guide instead of substituting the population estimate, which our measurements show is the safer choice. On paired data built from real EEG and EOG, subject-calibrated guidance improved reconstruction for every development participant and for nearly every participant in a held-out cohort evaluated once under a frozen protocol. On natural recordings it removed more ocular signal at comparable preservation of the recording than population calibration. Sampling the propagation matrix together with the diffusion trajectories yields predictive intervals that reach nominal coverage with no loss in probabilistic accuracy.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health informatics**; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Additional Key Words and Phrases: EEG denoising, ocular artifacts, electrooculography, conditional diffusion, subject calibration, predictive intervals

1 Introduction

Electroencephalography (EEG) records neural activity at the scalp with millisecond resolution, but the same electrodes also pick up the potentials generated by blinks and eye movements. These ocular artifacts are often larger than the neural activity of interest, spread across many channels, and overlap it in time and frequency [Croft and Barry 2000]. Rejecting every affected segment can discard a large fraction of a recording, while aggressive correction attenuates the event-related potentials and oscillations the recording was made to capture. Ocular-artifact removal is therefore a restoration problem: reduce the contribution of the eyes while leaving the neural signal intact. Classical regression solves it with an explicit physical model. When electrooculography (EOG) is recorded alongside EEG, the propagation of ocular potentials to each scalp channel can be estimated and the predicted contribution subtracted [Gratton et al. 1983; Semlitsch et al. 1986]. Blind source separation and artifact subspace reconstruction remove the need for an EOG channel [Chang et al. 2020; Jung et al. 2000], and supervised networks trained on paired corpora learn far more flexible corrections than a fixed linear subtraction [Yu et al. 2022; Zhang et al. 2021]. Each family, however, fixes one side of the problem. Regression models the artifact pathway of one person but knows nothing about the structure of clean EEG. A trained network knows a great deal about clean EEG but averages the artifact pathway over its training population, and neither a subject label nor an embedding of the corrupted recording tells it how a particular person's eye activity actually reaches the electrodes.

The premise of this paper is that the subject-specific knowledge a denoiser needs is not an identity code. It is the operator that carries the subject: the EOG-to-EEG propagation matrix, which depends on anatomy, electrode placement, and session conditions, and which can be measured directly from about two minutes of simultaneous EEG and EOG [Gratton et al. 1983; Kobler et al.

2020]. We therefore split the denoiser into a part that is shared and a part that is measured. A conditional diffusion model, trained once on a population, represents the distribution of clean EEG [Ho et al. 2020; Huang et al. 2025]. A short calibration segment, recorded before the data to be cleaned, yields a ridge estimate of the user’s propagation matrix. Empirical Bayes shrinkage pulls uncertain entries of this estimate toward the population value, and a reliability criterion decides whether the calibration can be used at all. At evaluation time the recorded EOG, passed through the calibrated matrix, becomes an explicit estimate of the ocular contribution that guides the diffusion model, and the same calibration supplies a per-channel feature vector that conditions the correction. Because the propagation estimate carries a known uncertainty, we sample it jointly with the diffusion trajectories, which turns the sampler into a source of predictive intervals whose width reflects both the model and the calibration.

Seen as a system, the method is a small adaptation loop rather than a fixed network: it measures the user’s artifact pathway, checks the measurement, acts on it, and reports its remaining uncertainty. Each stage has a defined failure behavior. If the calibration segment is too short or unstable, the guide is withheld and restoration proceeds without subject-specific input, a fail-safe we adopted after measuring the alternative and finding that substituting a population matrix subtracts the wrong signal. Adapting to a new user requires no retraining and no gradient access, only a new calibration segment, and our measurements show that this dedicated calibration cannot be replaced by adapting the propagation estimate on-line during use. The predictive intervals close the loop by exposing, per channel and time point, how much the system trusts its own output.

We evaluate the method on multimodal recordings with 46 EEG channels and two bipolar EOG derivations, using paired episodes built from real EEG and real ocular waveforms for reconstruction error, and natural recordings for attenuation and preservation. On fifteen development participants, subject-calibrated guidance improved the relative root-mean-square error over an unguided control for every participant, and over population calibration for fourteen of fifteen. Eight further participants were held out until the models and the analysis were frozen and then evaluated once; the improvement reproduced at nearly identical magnitude. On natural recordings the calibrated system attenuated substantially more ocular signal than population calibration while preserving more of the recording. The predictive intervals reach nominal 80% and 90% coverage with a continuous ranked probability score slightly better than that of the uncalibrated sample distribution, so the calibration of uncertainty costs nothing in sharpness, and the sample-based intervals compare favorably with a deterministic ensemble. A supporting experiment on a second dataset with 259 training participants shows that the same participant-matched calibration information remains beneficial when delivered through a learned representation, and that the benefit neither grows nor shrinks with training-population size, which is consistent with our central premise: the useful subject information lives in the measured operator, not in the size of the population model.

This paper makes four contributions. First, a subject-calibrated EOG-guided diffusion method that combines closed-form propagation calibration, empirical Bayes shrinkage, and a reliability criterion with a fail-safe fallback, on top of a population diffusion model that is never retrained per user. Second, a controlled evaluation, with population, mismatched, unguided, and temporally shuffled comparisons, that locates the improvement in the user’s own propagation estimate, together with a single-pass held-out evaluation that reproduces it. Third, natural-recording and deployment measurements, including the fallback comparison and an on-line adaptation comparison, that characterize how the system behaves when calibration is unreliable or unavailable. Fourth, calibration-aware predictive intervals that reach nominal coverage at no cost in probabilistic accuracy.

2 Related Work

Regression-based ocular correction estimates how eye potentials propagate to each scalp channel and subtracts the predicted contribution [Gratton et al. 1983; Semlitsch et al. 1986]. Its correction coefficients have a direct physical reading, and later work showed that eye-related subspaces estimated once from a calibration recording can be applied on-line with matrix operations [Kobler et al. 2020]. Blind source separation and artifact subspace reconstruction instead exploit the spatial structure of multichannel EEG and need no EOG channel [Chang et al. 2020; Jung et al. 2000]. This literature established both the value of a calibration segment and the linear propagation model we build on, but it treats clean EEG only implicitly, as whatever remains after subtraction. Our method keeps the calibrated propagation model of this lineage and adds what it lacks, a learned distribution of clean EEG that decides how the subtraction should be corrected.

Supervised networks trained on semi-simulated corpora such as EEGdenoiseNet learn flexible mappings from contaminated to clean signals [Chuang et al. 2022; Lai et al. 2024; Sun et al. 2020; Yu et al. 2022; Zhang et al. 2021], and calibration-free models extend this to unseen montages [Marcos-Martinez et al. 2025]. In these models participant variation is absorbed into shared parameters. The closest prior work to ours in intent is the domain-specific denoising diffusion model of Duan et al. [2023], which makes diffusion-based EEG denoising subject-aware by learning subject-specific noise distributions inside the network, so that the subjects to be denoised participate in training. We share the goal and the diffusion backbone but place the subject information elsewhere: the network stays subject-independent, and subject specificity enters through a closed-form propagation estimate from the evaluated user's own calibration segment. The two designs answer different deployment questions. Learning subject structure into the weights suits a fixed user pool; measuring it at calibration time extends to users the network has never seen, which is the setting we evaluate.

Diffusion models learn to invert a prescribed noising process [Ho et al. 2020; Song et al. 2021b] and, conditioned on a degraded observation, restore images and audio [Lemercier et al. 2023; Saharia et al. 2022]; EEGDfus and D4PM apply conditional diffusion to EEG artifact removal [Huang et al. 2025; Shao et al. 2025]. When the degradation operator is known, restoration methods such as DDRM and diffusion posterior sampling use it explicitly during sampling [Chung et al. 2023b; Kawar et al. 2022]; when it is unknown, blind approaches estimate operator and signal jointly [Chung et al. 2023a; Murata et al. 2023]. Our setting sits between these cases. The operator is unknown a priori but measurable: a calibration segment provides it in closed form before sampling begins, with an uncertainty that we propagate into the sample distribution. We evaluate the resulting intervals by empirical coverage and the continuous ranked probability score [Gneiting and Raftery 2007], rather than treating sampler variability as calibrated uncertainty by construction.

3 Methodology

3.1 Calibrating the propagation matrix

For a participant-session-task cell i , let $\mathbf{y} \in \mathbb{R}^{C \times T}$ denote the observed EEG, $\mathbf{x}$ the underlying neural signal, and $\mathbf{e} \in \mathbb{R}^{R \times T}$ the simultaneously recorded EOG. Over the interval of interest we use the additive model

$$\mathbf{y} = \mathbf{x} + \mathbf{A}_i \mathbf{e} + \boldsymbol{\varepsilon}, \quad (1)$$

where $\mathbf{A}_i \in \mathbb{R}^{C \times R}$ is the EOG-to-EEG propagation matrix and $\boldsymbol{\varepsilon}$ collects residual activity outside the ocular model. This is the operational model of classical EOG regression [Gratton et al. 1983]: not a complete physiological forward model, but a description whose single unknown, $\mathbf{A}_i$, is exactly the quantity that differs across people and sessions.

The calibration segment is a 120-s prefix of the recording containing paired EEG and EOG, temporally disjoint from everything that is later denoised. Each EOG channel is centered by its

calibration median and scaled by 1.4826 times its median absolute deviation with a 10^{-6} lower bound; EEG channels use scales estimated from the training fold. After temporal centering, ridge regression gives

$$\widehat{\mathbf{A}}_i = \mathbf{Y}_c \mathbf{E}_c^\top (\mathbf{E}_c \mathbf{E}_c^\top + r \mathbf{I})^{-1}, \quad r = 0.05 \max \left\{ \frac{\text{tr}(\mathbf{E}_c \mathbf{E}_c^\top)}{R}, \epsilon \right\}, \quad (2)$$

with ϵ the machine precision. A 120-s ridge estimate is still noisy, so we shrink it toward a population matrix $\mathbf{A}_{\text{pop}}$, the mean of training-participant estimates from the same session–task cell with the evaluated participant excluded:

$$\widetilde{\mathbf{A}}_i = \mathbf{A}_{\text{pop}} + \lambda_i (\widehat{\mathbf{A}}_i - \mathbf{A}_{\text{pop}}), \quad 0 \leq \lambda_i \leq 1. \quad (3)$$

The shrinkage weight is an empirical Bayes reliability ratio. Dividing the calibration segment into four consecutive 30-s blocks, let τ^2 be the mean squared deviation of training participants' full estimates from $\mathbf{A}_{\text{pop}}$, across participants and matrix entries, and let v_i be the mean squared deviation of participant i 's block estimates from their full estimate. Then $\lambda_i = \tau^2 / (\tau^2 + v_i/4)$, clipped to $[0, 1]$: a calibration whose blocks agree earns its individual estimate, one whose blocks disagree is pulled to the population. The computation uses only calibration EEG and EOG, never a clean target or a participant identifier.

Shrinkage handles graded unreliability; a separate criterion handles outright failure. If less than 60 s of calibration is available, or v_i exceeds the 95th percentile of training-fold values, the calibration is rejected. A rejected calibration does not fall back to the population matrix. Instead, the artifact guide defined below is withheld entirely and the calibration features revert to their population values, so the system denoises without any subject-specific input for that cell. We adopted this fail-safe after measuring both options: subtracting a population-propagated artifact estimate from a person it does not fit removes real signal, and Section 5 quantifies how much safer withholding the guide is on exactly the cells the criterion catches.

3.2 Guided diffusion restoration

At evaluation time, the recorded EOG and the calibrated matrix form the artifact guide $\mathbf{a}_i = \widetilde{\mathbf{A}}_i \mathbf{e}$, an explicit waveform estimate of the ocular contribution in Equation 1. The EEG model that consumes it is shared across all participants: a population-trained one-dimensional U-Net receives the diffusion state $\mathbf{x}_t$, the contaminated EEG $\mathbf{y}$, the guide $\mathbf{a}_i$, the diffusion time t , and calibration features $\mathbf{c}_i \in \mathbb{R}^{C \times 53}$ holding, per channel, the two propagation coefficients, their log norm, four calibration-quality variables (vertical and horizontal EOG magnitude, ridge fit, and condition number), and a 46-dimensional channel indicator. A row encoder and a one-dimensional spatial convolution reduce these features to a 128-dimensional context and channelwise modulation weights. The clean estimate is parameterized around EOG regression,

$$\widehat{\mathbf{x}}_0 = (\mathbf{y} - \mathbf{a}_i) + \Delta_{\text{pop},\theta}(\mathbf{x}_t, \mathbf{y}, \mathbf{a}_i, t) + \Delta_{\text{cal},\theta}(\mathbf{x}_t, \mathbf{y}, \mathbf{a}_i, t, \mathbf{c}_i), \quad (4)$$

so the network starts from the classical subtraction and learns two corrections: $\Delta_{\text{pop},\theta}$ models EEG structure shared across participants, and $\Delta_{\text{cal},\theta}$ lets the correction depend on the calibrated propagation. Both are trained jointly with a Smooth L1 loss against reference clean EEG; sampling uses deterministic DDIM updates [Song et al. 2021a]. The role of the diffusion model is to represent a population distribution of plausible clean waveforms conditioned on the observation and its calibrated ocular component. The same parameters serve every user and every operating mode, including the fallback: withholding the guide simply sets $\mathbf{a}_i = \mathbf{0}$ in Equation 4, a mode the network has seen during training. No per-user retraining occurs anywhere in the system. Figure 1 summarizes the flow.

Figure design placeholder. Method overview (design). One left-to-right system diagram, three stages. Stage 1, calibration: a timeline showing the 120-s calibration prefix disjoint from later evaluation windows; simultaneous EEG and EOG feed ridge regression (Eq. 2) and empirical Bayes shrinkage (Eq. 3) to produce $\tilde{A}_i$; a small branch shows the reliability criterion routing rejected calibrations to the fallback (guide withheld). Stage 2, restoration: evaluation EOG passes through $\tilde{A}_i$ to form the guide a_i , entering the shared U-Net inside the DDIM loop together with y and the calibration features; the output decomposition of Eq. 4 is drawn as subtraction plus two learned corrections. Stage 3, uncertainty: K trajectories with sampled propagation matrices (Eq. 5) produce a band around the restored waveform. Blue for shared population components, orange for calibration-derived quantities, gray for the fallback path. Render from the actual pipeline; no synthetic waveforms or performance values.

Fig. 1. The system calibrates the propagation matrix from a separate segment, guides a shared diffusion model with the propagated EOG, withholds the guide when calibration fails, and reports predictive intervals that include propagation uncertainty.

3.3 Predictive intervals

The calibrated matrix is an estimate with known uncertainty, and the sampler can carry that uncertainty into its output. For interval construction, $K = 32$ reverse trajectories jointly sample the DDIM initial noise and an entrywise propagation matrix

$$A_{i,cr}^{(k)} \sim \mathcal{N}(\tilde{A}_{i,cr}, \sigma_{A_{i,cr}}^2), \quad \sigma_{A_{i,cr}}^2 = \left(\frac{1}{\tau_{cr}^2} + \frac{4}{v_{i,cr}} \right)^{-1}, \quad (5)$$

where τ_{cr}^2 and $v_{i,cr}$ are the entrywise counterparts of the scalar between-participant and within-calibration variances of Section 3.1. The trajectory mean is the point estimate. The empirical trajectory spread w_{chain} is combined with the variance the sampled matrices propagate into each channel, $w_{A_{i,c}}^2(t) = \sum_r \sigma_{A_{i,cr}}^2 e_r(t)^2$, giving

$$w^2 = w_{\text{chain}}^2 + w_A^2, \quad I_{1-\alpha} = \bar{x} \pm \tau_{-f} z_{1-\alpha/2} w, \quad (6)$$

where τ_{-f} is a scalar temperature chosen without the evaluated fold, as the smallest value on a grid from 0.50 to 6.00 in steps of 0.05 reaching 80% coverage on the remaining folds, and z_p is the standard-normal quantile. The interval $I_{1-\alpha}$ is evaluated empirically, by marginal coverage, width, the continuous ranked probability score, and risk-coverage curves, in Section 5.

4 Experiments

4.1 Data and evaluation protocol

The main experiments use multimodal recordings with synchronized EEG and EOG from a public brain-computer interface dataset [Guttmann-Flury et al. 2024, 2025], represented as 46 EEG channels and two bipolar EOG derivations (vertical and horizontal) resampled to 100 Hz, with 5.12-s model inputs (512 samples). Within each session-task cell, a continuous 120-s prefix supplies the calibration and all evaluation windows occur later in the record. Fifteen participants form the development cohort, split into five participant-held-out folds of nine training, three validation, and three test participants. Eight further participants form the held-out cohort: they were excluded from every modeling and analysis decision, and were evaluated exactly once, after the fold models, the population registry, and the comparison had been frozen, with no iteration on the outcome. The source study was approved by the Institutional Review Board of Shanghai Jiao Tong University

(IRB HRP E2021216I) and this work is a secondary analysis of the public, de-identified recordings [Guttmann-Flury et al. 2025].

Reconstruction accuracy is measured on paired episodes built entirely from real signals. A low-ocular-activity EEG segment serves as the reference $\mathbf{x}$, a separately sampled real ocular waveform is passed through the recipient's later-record propagation matrix (estimated from 150–270 s, never an input to any model) to form the contamination, and the sum is the observation. Reference, ocular waveform, and recipient calibration come from three different participants, so the construction isolates the propagation pathway while keeping real EEG and real EOG throughout. Injected amplitudes use base gains of 0.35, 0.70, or 1.15 with a uniform factor from 0.85 to 1.15, and 15% of episodes contain no injected artifact; eight episodes are evaluated per participant across the available session–task cells. The paired endpoint is the relative root-mean-square error, $\text{RRMSE} = \|\hat{\mathbf{x}} - \mathbf{x}\|_2 / \|\mathbf{x}\|_2$, computed per episode and averaged within participant; improvements are reported as RRMSE differences in which positive values favor the first-named method. Natural recordings have no clean target and are analyzed separately on four windows per cell placed from 300 s onward, with three endpoints: EOG attenuation in decibels, the reduction of EOG–EEG coherence, and low-EOG retention, the proportion of the recorded waveform preserved in segments with little EOG activity. Paired and natural outcomes are never combined into one score.

A supporting experiment uses EEGEyeNet [Kastrati et al. 2021] to test whether the benefit of matched calibration depends on its matrix interface or on the information it conveys. A compact deterministic network receives an injected 46-channel episode and one of three calibration representations: participant-matched, zero, or mismatched. For each training-pool size from 30 to 259 participants the network is trained for 20,000 AdamW updates, then frozen, and a 32-dimensional representation is fitted per evaluation participant against paired clean calibration targets and tested on disjoint query episodes. Because the representation is fitted with clean targets, this experiment upper-bounds what calibration information can deliver through a learned interface; it is not part of the main method.

4.2 Comparison conditions

All diffusion variants share one population network and identical evaluation instances; they differ only in what the guide and calibration features contain (Table 1). *Matched calibration* is the proposed mode: the guide is the evaluation EOG propagated through the user's own shrunk matrix. *Population calibration* replaces the matrix by the recipient-excluded pooled estimate, testing whether an individual estimate is needed at all. *Mismatched calibration* uses another participant's matrix from the same session–task stratum, testing specificity. The *unguided control* withholds the waveform guide while keeping the calibration features, isolating the contribution of the explicit guide; it is also the operating mode the fallback of Section 3.1 reverts to. *Shuffled EOG* destroys the temporal alignment between EOG and EEG. Outside the diffusion family, we evaluate raw EEG, linear EOG regression from the same calibration [Gratton et al. 1983], a one-step deterministic network with the same inputs, an on-line variant in which the propagation matrix is tracked by recursive least squares during evaluation instead of being calibrated beforehand, and, for interval quality, an ensemble of the deterministic network.

4.3 Implementation and statistical analysis

The shared network is a four-scale one-dimensional U-Net (channel widths 32–128, four-head self-attention at the bottleneck, 128-dimensional calibration context, 965,085 parameters) predicting clean EEG over 1,000 linear diffusion levels ($\beta_1 = 10^{-4}$, $\beta_{1000} = 0.02$). Each fold and seed trains for 80,000 AdamW updates (batch 8, learning rate 10^{-4} , weight decay 10^{-4} , gradient clipping 1.0, exponential moving average 0.999). During training the guide is dropped with probability 0.30 and

Table 1. Comparison conditions and the information each receives. All diffusion modes use the same trained population network.

Condition	Propagation source	Guide / features	Question answered
Matched calibration	Own calibration segment	Propagated EOG; own features	Proposed method
Population calibration	Recipient-excluded pool	Propagated EOG; population features	Is the individual estimate needed?
Mismatched calibration	Another participant	Propagated EOG; donor features	Is the estimate specific to its owner?
Unguided control	Own calibration segment	No guide; own features	What does the explicit guide add?
Shuffled EOG	Own calibration segment	Misaligned guide; own features	Does timing matter?
Linear EOG regression	Own calibration segment	Direct subtraction	Classical reference
Deterministic network	Own calibration segment	One-step reconstruction	Learned point-estimate reference
On-line adaptation	Recursive least squares during use	Propagated EOG	Can calibration be replaced on-line?
Deterministic ensemble	Own calibration segment	Ensemble spread	Interval-quality reference

the calibration features are replaced by their population values with probability 0.20, so the unguided and fallback modes are in-distribution at evaluation time. Development point estimates use five folds, three seeds, and 50 deterministic DDIM steps with initial noise shared across conditions. The held-out evaluation uses one frozen model per fold, a population registry built from the 15 development participants, and the mean of the five fold predictions with a common noise draw per episode. The interval analysis uses $K = 32$ stochastic trajectories per development fold–seed cell with the temperature chosen without the evaluated fold. The paired comparisons reported in Section 5 were run with an earlier version of the rejection rule that substituted population calibration instead of withholding the guide; the two rules act identically on every cell whose calibration is accepted, and rerunning the development comparison under the final rule left the headline effect unchanged (Section 5.1).

All contrasts are aggregated within participant first: windows, protocols, seeds, and conditions are averaged per participant, and we report participant-level means, medians, counts of positive effects, and 95% percentile bootstrap intervals from 5,000 participant resamples. Interval coverage is computed over channel–time elements within each fold–seed cell and averaged over the 15 cells; the continuous ranked probability score and risk–coverage area are aggregated over episodes. Code, configurations, and stored evaluator outputs will be released on publication.

5 Results

5.1 Calibrated guidance improves paired restoration

On the fifteen development participants, matched calibration reached a participant-level RRMSE of 0.4310, against 0.5738 for the unguided control and 0.6511 for population calibration (Table 2). The improvement over the unguided control, 0.1428 (95% bootstrap interval 0.1061–0.1834), was positive for all fifteen participants and isolates what the explicit guide contributes beyond the shared network and calibration features. The improvement over population calibration, 0.2202

Table 2. Participant-level paired restoration. Lower RRMSE is better; improvements are participant-level mean differences with 95% bootstrap intervals. The two cohorts are never pooled.

Cohort	Condition	RRMSE	Comparison	Improvement (95% interval)	Positive
Development ($n = 15$)	Raw EEG	0.6076	–	–	–
	Population calibration	0.6511	Matched vs population	0.2202 (0.0929–0.4025)	14/15
	Unguided control	0.5738	Matched vs unguided	0.1428 (0.1061–0.1834)	15/15
	Matched calibration	0.4310	–	–	–
Held-out ($n = 8$)	Raw EEG	0.7753	–	–	–
	Population calibration	0.7661	–	–	–
	Unguided control	0.6888	Matched vs unguided	0.1537 (0.0725–0.2504)	7/8
	Matched calibration	0.5351	–	–	–

(0.0929–0.4025), was positive for fourteen of fifteen. Population calibration is in fact worse than raw EEG (0.6511 versus 0.6076), and the reason is instructive: an averaged propagation matrix applied to a person it does not fit subtracts activity that was never artifact. A population estimate is therefore useful as a shrinkage target, where it stabilizes an individual estimate, and harmful as a substitute, where it drives a subtraction of its own. This observation shaped the fallback of Section 3.1.

The fallback choice was then measured directly. The reliability criterion rejected the calibration in four development cells. On the subtraction pathway, replacing the population-substituted subtraction by no subtraction improved exactly those cells by 0.2537 (0.0070–0.5005) and the pooled error from 0.4340 to 0.4227; within the diffusion system the same change gave 0.0614 (–0.0258–0.1919) on the four cells, a positive point estimate whose interval spans zero at $n = 4$ because the learned corrections partially absorb a bad guide. On natural recordings from the rejected cells, withholding the guide raised retention from 0.917 to 0.984 while attenuation fell from 0.41 to 0.13 dB: the system stops subtracting where it cannot trust its calibration. Rerunning the full development comparison under this final rule left the headline effect unchanged, 0.1470 (0.1121–0.1871) with fifteen of fifteen positive; the two rules coincide on every cell whose calibration is accepted.

The eight held-out participants were then evaluated once, with frozen models, a frozen population registry, and the comparison fixed in advance. Matched calibration reached 0.5351 against 0.6888 for the unguided control, an improvement of 0.1537 (0.0725–0.2504) with seven of eight participants positive, close to the development effect despite the higher absolute error of this cohort. The comparison floor is itself demanding: on these participants the unguided control already improved on raw EEG by 0.086, so the calibrated gain sits on top of everything the population model and calibration features deliver. Figure 2 shows the paired waveforms and participant-level contrasts.

Against classical and deterministic references in the development protocol, matched diffusion (0.4310) and calibrated linear regression (0.4353) sit close together, with the one-step deterministic network at 0.4968; when both learned estimators are retrained with population calibration only, the deterministic network is ahead (0.5043 versus 0.5260). The point-estimate gain of this paper therefore comes from the calibration, not from the sampler: the propagation matrix that powers

Figure design placeholder. Paired denoising and participant effects (design). Panel A: a waveform strip for one development participant showing, on shared time and amplitude axes, the recorded EOG, the contaminated EEG, the paired reference, linear regression, the unguided control, and matched calibration for three fixed scalp channels; select the lowest-identifier participant with complete outputs and the earliest eligible ocular event, and state the identifiers in the final caption. Panel B: paired per-participant RRMSE lines from unguided to matched for all fifteen development participants. Panel C: the same for all eight held-out participants. Panel D: the two cohort-level improvements with bootstrap intervals on separate axes. All waveforms and points from stored evaluator outputs; nothing synthesized.

Fig. 2. Planned figure: what calibrated guidance changes in the waveform, and the per-participant effects behind Table 2.

Table 3. Natural-recording operating points, development cohort. Natural recordings have no clean target; higher attenuation and higher retention are both preferred.

Condition	EOG attenuation (dB)	Low-EOG retention
Unguided control	0.277	0.921
Population calibration	0.909	0.773
Matched calibration	2.463	0.843

the diffusion guide powers linear regression almost as well. What the sampler adds is a conditional distribution over clean waveforms, and Section 5.3 shows that this distribution outperforms a deterministic ensemble where distributions matter.

5.2 Natural recordings, specificity, and on-line adaptation

On natural development recordings, matched calibration attenuated 2.463 dB of EOG-coherent signal at a low-EOG retention of 0.843; population calibration attenuated 0.909 dB at retention 0.773, and the unguided control preserved the most (0.921) while attenuating the least (0.277 dB) (Table 3). Matched calibration thus dominates population calibration on both axes at once, gaining 1.554 dB of attenuation (0.969–2.211; 14/15 participants), 0.0707 of retention (0.0382–0.1067; 13/15), and 0.0849 of EOG–EEG coherence reduction (0.0535–0.1211; 14/15). The paired-error gain is not obtained by subtracting more everywhere; it is obtained by subtracting the right thing.

The controls pin down what the calibration is and is not. Substituting another participant’s matrix degrades reconstruction, and destroying the temporal alignment of the EOG removes most of the guide’s value, so the benefit rides on the specific propagation of the specific person, delivered at the right time. The reliability criterion, however, is a check on estimation quality, not on ownership: a stable matrix from the wrong person passes it, and the accepted mismatched condition remains 0.0384 worse than population calibration (0.0089–0.0697). The system accordingly assumes that calibration and evaluation data come from the same person and session, which is how the acquisition protocol is designed.

Figure design placeholder. Natural behavior, specificity, and adaptation (design). Panel A: operating points with low-EOG retention on the horizontal axis and EOG attenuation on the vertical axis for the unguided control, population, matched, mismatched, and shuffled-EOG conditions; matched should visibly dominate population on both axes. Panel B: per-participant matched-minus-population differences for attenuation, retention, and coherence reduction with bootstrap intervals. Panel C: RRMSE of the recursive least-squares tracker against adaptation time (10, 30, 60, 120, 240 s) as a curve, with the static calibrated value as a horizontal line. All values from stored evaluator outputs.

Fig. 3. Planned figure: matched calibration removes more while preserving more, the benefit is owner- and time-specific, and on-line adaptation does not reach the calibrated operating point.

Table 4. Predictive-interval quality on development data ($K = 32$). Coverage closer to nominal, lower CRPS, and lower risk-coverage area are better.

Interval policy	Coverage 50/80/90%	CRPS	Risk-coverage
Raw samples	0.302/0.479/0.558	0.1516	0.0520
Temperature only	0.625/0.802/0.851	0.1524	0.0523
Propagation + temperature	0.622/ 0.802/0.853	0.1503	0.0504

Calibration also cannot be replaced by adapting during use. A recursive least-squares tracker that updates the propagation estimate causally during evaluation reached RRMSE 0.944 after 10 s of adaptation, 0.654 after 30 s, 0.626 after 60 s, 0.601 after 120 s, and 1.051 after 240 s, never approaching the static calibrated value of 0.431; initializing the tracker wrongly made no lasting difference, since it converged to the same level within about 30 s. Evaluation-time data is dominated by exactly the artifacts being removed, which makes it poor material for estimating the artifact pathway. A two-minute dedicated calibration buys what continuous adaptation does not, which is why the system's adaptation loop runs through recalibration rather than through on-line updates. Figure 3 collects the natural operating points, the specificity controls, and the adaptation comparison.

5.3 Predictive intervals: calibrated coverage at no cost in sharpness

The raw distribution of the 32 trajectories is too narrow: empirical 50/80/90% coverages are 0.302, 0.479, and 0.558, with a continuous ranked probability score (CRPS) of 0.1516 (Table 4). A fold-held-out scalar temperature repairs coverage (0.625/0.802/0.851) at CRPS 0.1524. Adding the propagation-matrix uncertainty of Equation 5 and then temperature-scaling reaches 0.622/0.802/0.853 at CRPS 0.1503, with the risk-coverage area improving from 0.0520 to 0.0504. The 80% and 90% levels are within 0.05 of nominal, and the calibrated intervals are not blunter than the raw ones, they are sharper: because the raw samples were underdispersed, widening them toward the truth lowers CRPS, and the physically motivated width does it better than temperature alone, with a lower CRPS (0.1503 versus 0.1524) and a smaller required temperature (2.30–2.55 versus 2.90–3.20). The propagation term carries real width exactly where the calibration is uncertain. The 50% interval remains conservative (0.622), the one nominal level a single scalar does not align.

Figure design placeholder. Intervals and calibration information (design). Panel A: nominal versus empirical coverage for the three interval policies of Table 4 with the identity line. Panel B: CRPS and risk-coverage area as grouped bars for the three policies, and, on a separate clearly labeled axis pair, the diffusion-versus-deterministic-ensemble comparison. Panel C: matched-versus-zero representation gain against training-pool size (30, 60, 120, 200, 259) with bootstrap intervals, flat by Section 5.4. Panel D: matched, zero, and mismatched representation contrasts at the largest pool, labeled as the plausibility-restricted analysis with 14 participants. All values from stored aggregates; no invented bands.

Fig. 4. Planned figure: interval calibration at no sharpness cost, the sampler's advantage over a deterministic ensemble, and the flat, interface-independent value of matched calibration.

This is also where the diffusion sampler earns its place over deterministic alternatives. In a separate analysis of the same development episodes (with a different aggregation, so absolute values are not comparable across analyses), the diffusion sample distribution beat an ensemble of the deterministic network on CRPS (0.1529 versus 0.1548) and on risk-coverage area (0.0920 versus 0.1129), and after conformal adjustment its 80% held-out-fold coverage was 0.765 against 0.652 for the ensemble. Combined with Section 5.1, the division of labor is clear: the calibrated operator delivers the point-estimate gain, and the diffusion model turns that calibrated knowledge into the better predictive distribution. Figure 4 specifies the interval and representation visualizations.

5.4 Where the calibration information lives

The supporting EEGEyeNet experiment asks whether the benefit of matched calibration is tied to the propagation-matrix interface or to the information the calibration carries. Delivered instead through a support-fitted 32-dimensional representation conditioning a deterministic network, participant-matched calibration reduced RRMSE by 0.0608 relative to a zero representation (0.0406–0.0835; 14 of 14 participants) in the plausibility-restricted analysis, and a mismatched representation was harmful, with a matched-minus-mismatched separation of 0.1623 (0.1314–0.1946; 14 of 14). The complete 15-participant analysis was inconclusive (–1.428, interval –4.415–0.076) because one participant's support episodes carried an artifact-to-signal ratio of 167 against 0.33–4.02 for all others; restricting injected ratios to [0.1, 20] gives the analysis above. Since the representation is fitted with paired clean calibration targets, these numbers upper-bound a learned calibration interface rather than certify one. Two conclusions still stand. The value of matched calibration is information, not interface: it survives a change of both model class and delivery mechanism. And it is flat in training-population size, changing by –0.0224 (–0.0491–0.0158) from 30 to 259 training participants: enlarging the population model neither creates nor removes the need to calibrate the individual, which is the premise of this paper measured at scale.

6 Conclusion

Subject awareness in EEG denoising does not require learning the subject into the network. The quantity that actually differs across people, the propagation of ocular potentials to the scalp, can be measured from two minutes of paired EEG and EOG, stabilized by shrinkage, checked for reliability, and handed to a population diffusion model as an explicit artifact estimate. On this

design the individual estimate improved reconstruction for every development participant and reproduced at magnitude in a single-pass held-out evaluation; on natural recordings it removed more ocular signal while preserving more of the recording than population calibration; and the same calibration uncertainty, propagated into the sampler, produced predictive intervals with nominal 80% and 90% coverage at a slightly better probabilistic score than the uncalibrated samples, ahead of a deterministic ensemble. The failure modes are measured parts of the system rather than caveats around it: a rejected calibration withholds the guide, because substituting a population estimate subtracts the wrong signal; a wrong person's calibration is harmful, so the protocol keeps calibration and use in the same session; and on-line adaptation during use never reaches the calibrated operating point, so adaptation runs through recalibration.

The evaluated scope is same-session ocular-artifact removal with recorded EOG and a compatible montage. Natural recordings supply attenuation and preservation measures, not a clean target; the interval analysis is a development-cohort result, with a held-out confirmation on reserved data planned as the next step; the held-out cohort has eight participants, mitigated by the frozen single-pass protocol rather than by sample size; and as a point estimator the calibrated diffusion model is on par with calibrated linear regression, so the case for the sampler rests on its predictive distribution. External EEG–EOG cohorts, non-ocular artifact families, and compute-matched distributional baselines are the natural extensions. Within the evaluated setting the conclusion is direct: the operator carries the subject, and calibrating it adapts a shared generative denoiser to a person the model has never seen.

References

- Chi-Yuan Chang, Sheng-Hsiou Hsu, Luca Pion-Tonachini, and Tzyy-Ping Jung. 2020. Evaluation of Artifact Subspace Reconstruction for Automatic Artifact Components Removal in Multi-Channel EEG Recordings. *IEEE Transactions on Biomedical Engineering* 67, 4 (2020), 1114–1121. doi:10.1109/TBME.2019.2930186
- Chun-Hsiang Chuang, Kong-Yi Chang, Chih-Sheng Huang, and Tzyy-Ping Jung. 2022. IC-U-Net: A U-Net-Based Denoising Autoencoder Using Mixtures of Independent Components for Automatic EEG Artifact Removal. *NeuroImage* 263 (2022), 119586. doi:10.1016/j.neuroimage.2022.119586
- Hyungjin Chung, Jeongsol Kim, Sehui Kim, and Jong Chul Ye. 2023a. Parallel Diffusion Models of Operator and Image for Blind Inverse Problems. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6059–6069. doi:10.1109/CVPR52729.2023.00587
- Hyungjin Chung, Jeongsol Kim, Michael T. McCann, Marc L. Klasky, and Jong Chul Ye. 2023b. Diffusion Posterior Sampling for General Noisy Inverse Problems. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=OnD9zGAGT0k>
- R. J. Croft and R. J. Barry. 2000. Removal of Ocular Artifact from the EEG: A Review. *Neurophysiologie Clinique/Clinical Neurophysiology* 30, 1 (2000), 5–19. doi:10.1016/S0987-7053(00)00055-1
- Yiqun Duan, Jinzhao Zhou, Zhen Wang, Yu-Cheng Chang, Yu-Kai Wang, and Chin-Teng Lin. 2023. Domain-Specific Denoising Diffusion Probabilistic Models for Brain Dynamics. Preprint; no confirmed archival venue. arXiv:2305.04200 [cs.HC] doi:10.48550/arXiv.2305.04200
- Tilmann Gneiting and Adrian E. Raftery. 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. *J. Amer. Statist. Assoc.* 102, 477 (2007), 359–378. doi:10.1198/016214506000001437
- Gabriele Gratton, Michael G. H. Coles, and Emanuel Donchin. 1983. A New Method for Off-Line Removal of Ocular Artifact. *Electroencephalography and Clinical Neurophysiology* 55, 4 (1983), 468–484. doi:10.1016/0013-4694(83)90135-9
- Eva Guttmann-Flury, Xinjun Sheng, and Xiangyang Zhu. 2024. Eye-BCI_multi_dataset. doi:10.7303/syn64005218
- Eva Guttmann-Flury, Xinjun Sheng, and Xiangyang Zhu. 2025. Dataset Combining EEG, Eye-Tracking, and High-Speed Video for Ocular Activity Analysis across BCI Paradigms. *Scientific Data* 12, 1 (2025), 587. doi:10.1038/s41597-025-04861-9
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems*, Vol. 33. 6840–6851. <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>
- Xiaoyang Huang, Chang Li, Aiping Liu, Ruobing Qian, and Xun Chen. 2025. EEGDfus: A Conditional Diffusion Model for Fine-Grained EEG Denoising. *IEEE Journal of Biomedical and Health Informatics* 29, 4 (2025), 2557–2569. doi:10.1109/JBHI.2024.3504716

- Tzyy-Ping Jung, Scott Makeig, Colin Humphries, Te-Won Lee, Martin J. McKeown, Vicente Iragui, and Terrence J. Sejnowski. 2000. Removing Electroencephalographic Artifacts by Blind Source Separation. *Psychophysiology* 37, 2 (2000), 163–178. doi:10.1111/1469-8986.3720163
- Ard Kastrati, Martyna Beata Plomecka, Damián Pascual, Lukas Wolf, Victor Gillioz, Roger Wattenhofer, and Nicolas Langer. 2021. EEGEyeNet: A Simultaneous Electroencephalography and Eye-Tracking Dataset and Benchmark for Eye Movement Prediction. arXiv:2111.05100 [cs.LG] doi:10.48550/arXiv.2111.05100
- Bahjat Kawar, Michael Elad, Stefano Ermon, and Jiaming Song. 2022. Denoising Diffusion Restoration Models. In *Advances in Neural Information Processing Systems*, Vol. 35. 23593–23606. doi:10.52202/068431-1714
- Reinmar J. Kobler, Andreea I. Sburla, Catarina Lopes-Dias, Andreas Schwarz, Masayuki Hirata, and Gernot R. Müller-Putz. 2020. Corneo-Retinal-Dipole and Eyelid-Related Eye Artifacts Can Be Corrected Offline and Online in Electroencephalographic and Magnetoencephalographic Signals. *NeuroImage* 218 (2020), 117000. doi:10.1016/j.neuroimage.2020.117000
- Pin-Hua Lai, Bo-Shan Wang, Wei-Chun Yang, Hsiang-Chieh Tsou, and Chun-Shu Wei. 2024. CLEEGN: A Convolutional Neural Network for Plug-and-Play Automatic EEG Reconstruction. arXiv v2; originally submitted in 2022. arXiv:2210.05988 [eess.SP] doi:10.48550/arXiv.2210.05988
- Jean-Marie Lemerrier, Julius Richter, Simon Welker, and Timo Gerkmann. 2023. StoRM: A Diffusion-Based Stochastic Regeneration Model for Speech Enhancement and Dereverberation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 31 (2023), 2724–2737. doi:10.1109/TASLP.2023.3296602
- Diego Marcos-Martínez, Sergio Pérez-Velasco, Victor Martínez-Cagigal, Eduardo Santamaría-Vázquez, and Roberto Hornero. 2025. Calibration-Free Ocular Artifact Reduction in EEG Signals Using a Montage-Independent Deep Learning Model. *Biomedical Signal Processing and Control* 110 (2025), 108147. doi:10.1016/j.bspc.2025.108147
- Naoki Murata, Koichi Saito, Chieh-Hsin Lai, Yuhta Takida, Toshimitsu Uesaka, Yuki Mitsufuji, and Stefano Ermon. 2023. GibbsDDRM: A Partially Collapsed Gibbs Sampler for Solving Blind Inverse Problems with Denoising Diffusion Restoration. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*. 25501–25522. <https://proceedings.mlr.press/v202/murata23a.html>
- Chitwan Saharia, William Chan, Huiwen Chang, Chris Lee, Jonathan Ho, Tim Salimans, David J. Fleet, and Mohammad Norouzi. 2022. Palette: Image-to-Image Diffusion Models. In *ACM SIGGRAPH 2022 Conference Proceedings*. 1–10. doi:10.1145/3528233.3530757
- Heribert V. Semlitsch, Peter Anderer, Peter Schuster, and Otto Presslich. 1986. A Solution for Reliable and Valid Reduction of Ocular Artifacts, Applied to the P300 ERP. *Psychophysiology* 23, 6 (1986), 695–703. doi:10.1111/j.1469-8986.1986.tb00696.x
- Feixue Shao, Xueyu Liu, Yongfei Wu, Jianbo Lu, Guiying Yan, and Weihua Yang. 2025. D4PM: A Dual-Branch Driven Denoising Diffusion Probabilistic Model with Joint Posterior Diffusion Sampling for EEG Artifacts Removal. Preprint. arXiv:2509.14302 [eess.IV] doi:10.48550/arXiv.2509.14302
- Jiaming Song, Chenlin Meng, and Stefano Ermon. 2021a. Denoising Diffusion Implicit Models. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=St1giarCHLP>
- Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021b. Score-Based Generative Modeling through Stochastic Differential Equations. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=PxTIG12RRHS>
- Weitong Sun, Yuping Su, Xia Wu, and Xiaojun Wu. 2020. A Novel End-to-End 1D-ResCNN Model to Remove Artifact from EEG Signals. *Neurocomputing* 404 (2020), 108–121. doi:10.1016/j.neucom.2020.04.029
- Junjie Yu, Chenyi Li, Xexin Lou, Chen Wei, and Quanying Liu. 2022. Embedding Decomposition for Artifacts Removal in EEG Signals. *Journal of Neural Engineering* 19, 2 (2022), 026052. doi:10.1088/1741-2552/ac63eb
- Haoming Zhang, Mingqi Zhao, Chen Wei, Dante Mantini, Zherui Li, and Quanying Liu. 2021. EEGdenoiseNet: A Benchmark Dataset for Deep Learning Solutions of EEG Denoising. *Journal of Neural Engineering* 18, 5 (2021), 056057. doi:10.1088/1741-2552/ac2bf8